Local Order of Strang Splitting for Finite-State Markov Generators: a Short Proof

1 Statement

For square matrices B, C of the same size put

$$A = B + C, \qquad S(h) = e^{hB/2} e^{hC} e^{hB/2}, \qquad P(h) = e^{hA}.$$

The splitting is *exact* if $S(h) = P(h)$ for all real h. Otherwise $p(B, C)$ denotes the order of vanishing of $S(h) - P(h)$ at $h = 0$:

$$S(h) - P(h) = h^p K + O(h^{p+1}), \qquad K \neq 0.$$

A *Markov generator* is a real matrix with nonnegative off-diagonal entries and zero row sums. For $n \geq 2$ let

$$M_n = \sup\{\, p(B, C) : B, C \text{ are } n \times n \text{ Markov generators with nonexact splitting} \,\}.$$

Theorem 1. *For every $n \geq 2$,*

$$2 \left\lfloor \frac{n}{2} \right\rfloor + 1 \;\leq\; M_n \;\leq\; \max\{3,\, (n-1)(n-2) + 1\}.$$

The lower bound is attained by generators whose off-diagonal entries are all strictly positive. In particular $M_2 = M_3 = 3$.

Throughout, $[U, V] = UV - VU$, $\mathrm{ad}_U V = [U, V]$, and we fix the notation

$$\boxed{D = B + 2C, \qquad T = \mathrm{ad}_B, \qquad Z = \mathrm{ad}_D\,.}$$

We use $e^U V e^{-U} = e^{\mathrm{ad}_U} V$ freely. Lemmas 1–3 hold for arbitrary complex square matrices.

2 The order criterion

Lemma 1 (Parity)**.** *If the splitting is nonexact, then $p(B, C)$ is odd.*

Proof. For small h the principal logarithm $\Lambda(h) = \log S(h)$ is analytic, and $S(-h) = S(h)^{-1}$ gives $\Lambda(-h) = -\Lambda(h)$. Since $S(h) = I + hA + O(h^2)$, we have $\Lambda(h) = hA + O(h^2)$. If $\Lambda(h) = hA$ identically near 0, then $S = P$ near 0 and hence everywhere by analyticity. Otherwise $\Lambda(h) = hA + h^q K + O(h^{q+1})$ with $K \neq 0$, and q is odd because Λ is odd. Both $\Lambda(h)$ and hA are $O(h)$, so

$$S(h) - P(h) = e^{\Lambda(h)} - e^{hA} = h^q K + O(h^{q+1}),$$

and $p = q$. □

Lemma 2 (Order criterion)**.** *The splitting is exact if and only if $Z\,T^s D = 0$ for every $s \ge 1$, that is, if and only if D commutes with $\mathrm{ad}_B^s D$ for all $s \ge 1$. Otherwise*

$$p(B,C) = 2 + \min\{\, s \ge 1 : Z\,T^s D \ne 0 \,\}.$$

In particular $p \ge 3$, and $p > 3$ if and only if $[D,[B,D]] = 0$.

Proof. Step 1: a defect function. Let $E = S - P$ and $L = S'S^{-1} - A$. Since $P' = AP$,

$$L = (E' - AE)\,S^{-1}. \tag{1}$$

Differentiating the three factors of S,

$$S'S^{-1} = \tfrac{1}{2}B + e^{hT/2}C + \tfrac{1}{2}\,e^{hT/2}e^{h\,\mathrm{ad}_C}B.$$

Put $R = e^{-hT/2}L$. Using $e^{-hT/2}B = B$,

$$R(h) = \tfrac{1}{2}\big(e^{h\,\mathrm{ad}_C} - I\big)B - \big(e^{-hT/2} - I\big)C, \qquad R'(h) = \tfrac{1}{2}\big(e^{-hT/2} - e^{h\,\mathrm{ad}_C}\big)Q, \quad Q = [B,C].$$

Step 2: the order of R' is $p-2$. If $E \equiv 0$ then $L \equiv 0$ by (1), so $R' \equiv 0$. Conversely, if $R' \equiv 0$ then $R \equiv R(0) = 0$, so $L \equiv 0$, so $S' = AS$ with $S(0) = I$, and $S = P$. If $E = h^p K + O(h^{p+1})$ with $K \ne 0$, then (1) gives $L = p\,h^{p-1}K + O(h^p)$, and R has the same leading term. Since $R(0) = 0$, this forces $p \ge 2$, and then $R' = p(p-1)h^{p-2}K + O(h^{p-1})$ vanishes to order exactly $p-2$.

Step 3: rewriting in terms of D. Since $C = (D - B)/2$, we have $\mathrm{ad}_C = \tfrac{1}{2}(Z - T)$ and $Q = \tfrac{1}{2}[B,D] = \tfrac{1}{2}TD$. Hence

$$R'(h) = \sum_{j \ge 0} \frac{h^j}{j!\,2^{j+2}}\,(b_j - a_j), \qquad a_j = (Z-T)^j TD, \quad b_j = (-T)^j TD.$$

So the splitting is exact iff $a_j = b_j$ for all j, and otherwise $p - 2 = \min\{j : a_j \ne b_j\}$.

Step 4: induction. Suppose $a_{j-1} = b_{j-1}$ for some $j \ge 1$. Then

$$a_j = (Z - T)\,b_{j-1} = (-1)^{j-1} Z\,T^j D + b_j.$$

As $a_0 = b_0$, it follows that $a_j = b_j$ for all $j < s^*$, where $s^* = \min\{s \ge 1 : ZT^sD \ne 0\}$, and $a_{s^*} \ne b_{s^*}$ if s^* is finite. Thus $p - 2 = s^*$, and the splitting is exact iff $s^* = \infty$. The last claim is the case $s = 1$: $ZTD = [D,[B,D]]$. □

Remark 1. Since $[D, e^{hB}De^{-hB}] = \sum_{s \ge 1} \frac{h^s}{s!} ZT^sD$, the criterion reads

$$p(B,C) = 2 + \mathrm{ord}_{h=0}\,\big[D,\ e^{hB}De^{-hB}\big],$$

and the splitting is exact iff D commutes with all its conjugates $e^{hB}De^{-hB}$. The leading error coefficient for $p = 3$ is $K = -\frac{1}{48}[D,[B,D]] = -\frac{1}{24}[B + 2C,[B,C]]$.

3 Upper bound

Lemma 3. *For $m \times m$ matrices with nonexact splitting, $p(B,C) \le m(m-1) + 1$.*

Proof. Let $W_j = \operatorname{span}\{T^s D : 1 \le s \le j\}$ and $W = \bigcup_j W_j$. If $T^{j+1}D \in W_j$ then $TW_j \subseteq W_j$ and $W = W_j$. Hence $\dim W_j = j$ until the chain stabilises, and $W = W_\ell$ with $\ell = \dim W$.

Every $T^s D$ with $s \ge 1$ lies in $\operatorname{im} T$, so $\ell \le \operatorname{rank} \operatorname{ad}_B = m^2 - \dim \ker \operatorname{ad}_B \le m^2 - m$. The last inequality holds because $\dim \ker \operatorname{ad}_B = m$ when B has distinct eigenvalues (the centraliser is then the algebra of polynomials in B), such B are dense, and $\operatorname{rank} \operatorname{ad}_B$ is lower semicontinuous in B.

If $ZT^s D = 0$ for $1 \le s \le \ell$, then Z vanishes on $W_\ell = W$ and the splitting is exact by Lemma 2. So a nonexact pair has $s^* \le \ell \le m(m-1)$, i.e. $p \le m(m-1) + 2$. This number is even and p is odd (Lemma 1). □

Proof of the upper bound in Theorem 1. Let B, C be $n \times n$ Markov generators with nonexact splitting and $p > 3$. By Lemma 2, $Z^2 B = -[D, [B, D]] = 0$.

B and C preserve the generalised eigenspaces of D. Write $\mathbb{C}^n = \bigoplus_\lambda E_\lambda$ with E_λ the generalised eigenspaces of D, and let $B_{\lambda\mu} : E_\mu \to E_\lambda$ be the blocks of B. With $D|_{E_\lambda} = \lambda + N_\lambda$, N_λ nilpotent,

$$(ZB)_{\lambda\mu} = \Phi(B_{\lambda\mu}), \qquad \Phi(X) = (\lambda - \mu)X + (N_\lambda X - X N_\mu).$$

Left multiplication by N_λ and right multiplication by N_μ are commuting nilpotent operators, so Φ is invertible when $\lambda \ne \mu$. Then $\Phi^2(B_{\lambda\mu}) = 0$ forces $B_{\lambda\mu} = 0$. So B, and hence $C = (D - B)/2$, is block diagonal.

The block at 0 is exact. $D = B + 2C$ is a Markov generator, so e^{tD} is a stochastic matrix for $t \ge 0$ and $\|e^{tD}\|_\infty = 1$. A Jordan block of size ≥ 2 at the eigenvalue 0 would make e^{tD} grow linearly in t. Hence $E_0 = \ker D$, and $E_0 \ni \mathbf{1}$ is nonzero. On E_0 we have $C = -B/2$, so B and C commute there and $S = P$ on E_0.

Conclusion. Let $W' = \bigoplus_{\lambda \ne 0} E_\lambda$, of dimension $m \le n - 1$. Since $S - P = 0 \oplus (S - P)|_{W'}$, the restricted pair is nonexact with the same order, and Lemma 3 gives $p \le m(m-1) + 1 \le (n-1)(n-2) + 1$. □

4 Lower bound

Lemma 4 (Shift–corner pair)**.** *Let $m \ge 2$, let E_{ij} be the matrix units, and set*

$$B = F = \sum_{i=1}^{m-1} E_{i+1,i}, \qquad D = E_{1m}, \qquad C = \tfrac{1}{2}(D - F).$$

Then $B + 2C = D$, the splitting is nonexact, and

$$p(B, C) = \begin{cases} m + 1, & m \text{ even}, \\ m + 2, & m \text{ odd}. \end{cases}$$

Proof. Left and right multiplication by F commute, and $F^a E_{1m} F^b = E_{1+a,\, m-b}$ for $0 \le a, b \le m-1$ (zero otherwise). Hence

$$T^s D = \sum_{\substack{a+b=s \\ 0 \le a, b \le m-1}} (-1)^b \binom{s}{b} E_{1+a,\, m-b}.$$

Also $ZE_{ij} = \delta_{mi} E_{1j} - \delta_{j1} E_{im}$, so Z annihilates $E_{1+a,m-b}$ unless $a = m-1$ or $b = m-1$. Therefore:

- $ZT^s D = 0$ for $1 \le s \le m - 2$;
- for $s = m - 1$ the terms $(a, b) = (m-1, 0)$ and $(0, m-1)$ give $ZT^{m-1}D = \big(1 + (-1)^m\big)D$;

- for $s = m$ and $m \geq 3$ the terms $(m-1,1)$ and $(1,m-1)$ give $ZT^mD = -m\,E_{1,m-1} + (-1)^m m\,E_{2,m}$.

If m is even, the first nonzero term is at $s = m-1$. If m is odd, it is at $s = m$. Apply Lemma 2. □

Lemma 5 (Markov embedding)**.** *Let B, C be real $m \times m$ matrices and $n = m+1$. There are $n \times n$ Markov generators B', C' with strictly positive off-diagonal entries whose splitting is exact iff that of B, C is, and has the same order otherwise.*

Proof. Let U be an $n \times m$ matrix whose columns are an orthonormal basis of $\mathbf{1}^{\perp}$, and $J = \frac{1}{n}\mathbf{1}\mathbf{1}^{\top}$. Then $U^{\top}U = I_m$, $U^{\top}\mathbf{1} = 0$, $UU^{\top} = I_n - J$. For $\lambda > 0$ put

$$B' = U(B - \lambda I_m)U^{\top} = UBU^{\top} + \lambda(J - I_n), \qquad C' = U(C - \lambda I_m)U^{\top}.$$

Row sums vanish because $U^{\top}\mathbf{1} = 0$. Each off-diagonal entry is an entry of $UBU^{\top}$ or $UCU^{\top}$ plus λ/n, hence positive for λ large.

Since $(UXU^{\top})(UYU^{\top}) = UXYU^{\top}$, we get $e^{UXU^{\top}} = J + Ue^{X}U^{\top}$, and $JU = 0$. Scalars commute with everything, so

$$e^{hB'/2}e^{hC'}e^{hB'/2} - e^{h(B'+C')} = e^{-2\lambda h}\,U\big(S(h) - P(h)\big)U^{\top}.$$

The factor $e^{-2\lambda h}$ never vanishes and $X \mapsto UXU^{\top}$ is injective, so exactness and the order of vanishing are preserved. □

Proof of the lower bound and of $M_2 = M_3 = 3$. For $n \geq 3$, apply Lemma 5 to the pair of Lemma 4 with $m = n-1$. The order is n if n is odd and $n+1$ if n is even, i.e. $2\lfloor n/2 \rfloor + 1$.

For $n = 2$ take

$$B = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad C = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix}, \qquad [D,[B,D]] = 16\begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \neq 0,$$

so $p = 3$ by Lemma 2.

For $n = 2, 3$ both bounds equal 3. □